Mutual Fund Analytics Platform

Blue stock Fintech Capstone Project

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Tools Used: Python, Pandas, NumPy, SQLite, SQL Alchemy, Jupyter Notebook, Power BI, Git, GitHub
Project Duration: Day 1 – Day 4
Report Version: Final Submission

Executive Summary

The Mutual Fund Analytics Platform was developed to analyze mutual fund performance, investor behaviour, portfolio risk, and industry trends using historical and transactional data. The project integrates data engineering, data analytics, risk analytics, SQL database design, and dashboard visualization into a single end-to-end analytics solution.

The platform processes ten mutual fund datasets containing fund metadata, NAV history, investor transactions, AUM information, portfolio holdings, and benchmark data. Through ETL pipelines, data quality validation, database modelling, and analytical reporting, the project provides actionable insights for investors, fund managers, and financial institutions.

Key outcomes include investor segmentation, risk-adjusted fund performance evaluation, Value-at-Risk analysis, SIP continuity monitoring, portfolio concentration assessment, and a rule-based fund recommendation engine.

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1. Project Objectives

Business Objectives

The project aims to:

- Analyze mutual fund performance across multiple schemes.
- Evaluate investor behaviour and transaction patterns.
- Measure portfolio risk using quantitative risk metrics.
- Monitor SIP continuity and investor retention.
- Compare fund diversification and concentration levels.
- Develop dashboards for executive decision-making.
- Create a recommendation framework based on investor risk profiles.

Technical Objectives

- Build ETL pipelines using Python.
- Design a star-schema SQLite data warehouse.
- Implement SQL analytics and reporting.
- Create Power BI dashboards.

- Perform risk analytics using historical returns.
- Document the complete analytics workflow.

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2. Data Sources

The project utilizes ten datasets:

Dataset	**Description**
Fund Master	Scheme metadata
NAV History	Historical daily NAV values
AUM by Fund House	Assets under management
SIP Inflows	Monthly SIP trends
Category Inflows	Category-level investment flows
Industry Folio Count	Industry growth metrics
Scheme Performance	Return and risk measures
Investor Transactions	Investor transaction history
Portfolio Holdings	Security and sector holdings
Benchmark Indices	Market benchmark data

External API

mfapi.in

Used for live NAV retrieval and validation of scheme performance.

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3. ETL Design & Data Engineering

Project Architecture

Raw Data → Data Cleaning → Validation → SQLite Warehouse → Analytics → Power BI Dashboard

Data Cleaning Activities

- Missing value handling
- Duplicate removal
- Data type standardization
- Transaction type normalization
- Date parsing and validation

- NAV validation
- Expense ratio validation
- KYC status verification

Data Quality Controls

- Referential integrity checks
- Row count validation
- Duplicate detection
- Range validation rules
- Null value analysis

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4. Database Design

Star Schema Overview

Dimension Tables

dim_fund

Contains fund metadata.

dim_date

Calendar dimension supporting time-series analysis.

Fact Tables

fact_nav

Daily NAV records.

fact_transactions

Investor transactions.

fact_performance

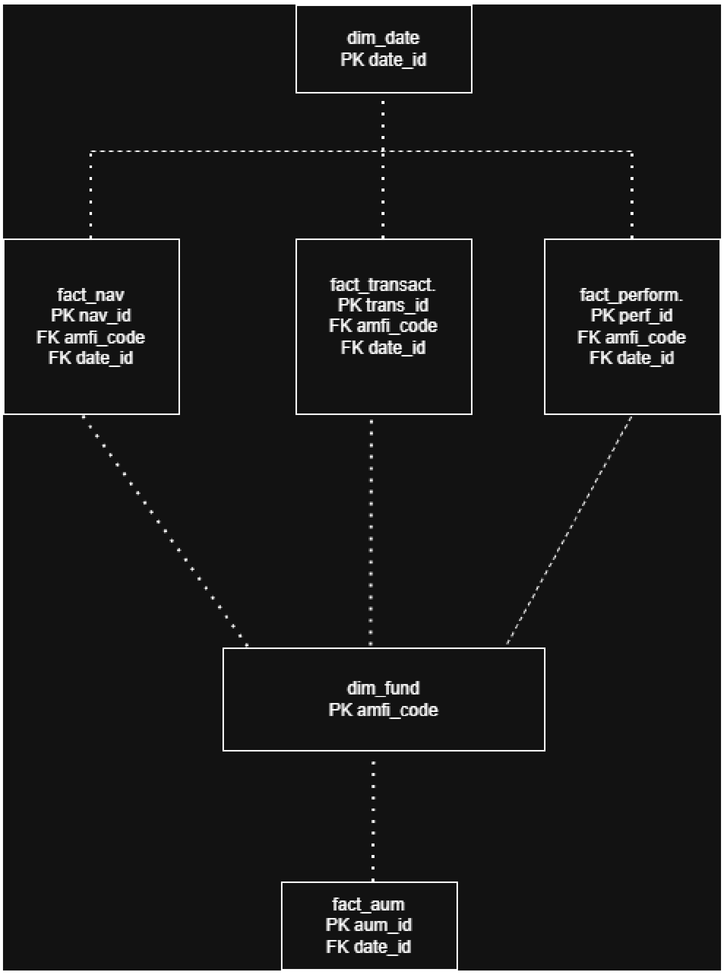
Risk and return metrics.

fact_aum

Fund house AUM records.

Database Benefits

- Faster analytical queries
- Reduced redundancy
- Better dashboard performance
- Improved scalability



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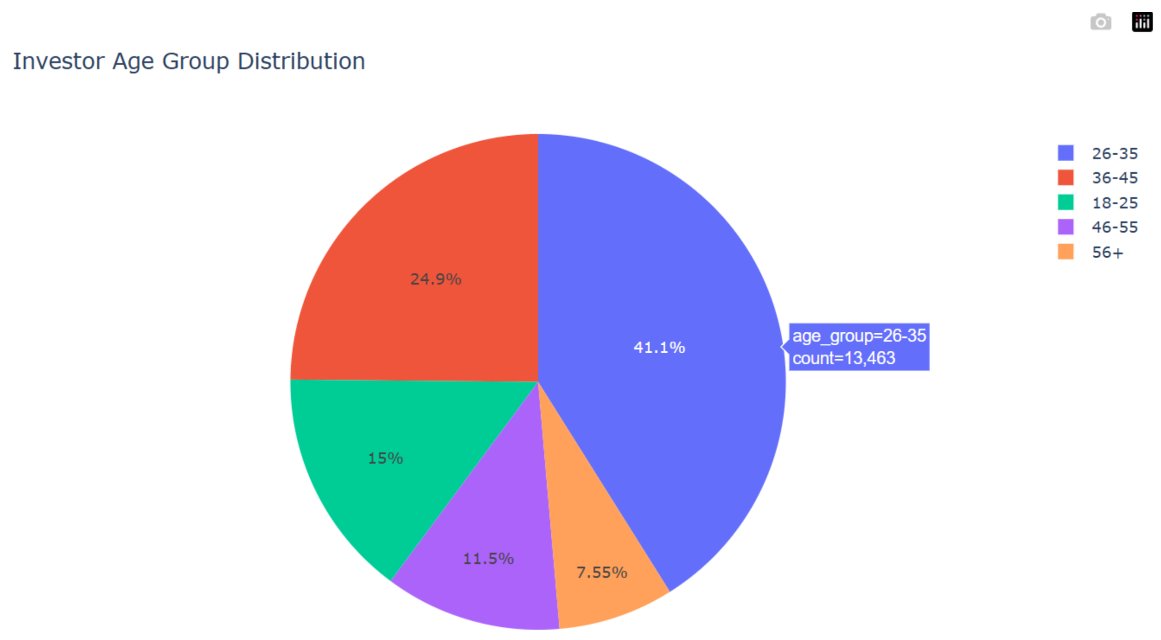
5. Exploratory Data Analysis

Investor Demographics

Age Group Distribution

Key findings:

- Majority of investors belong to the working-age population.
- Younger investors show increasing participation.



Gender Distribution

Key findings:

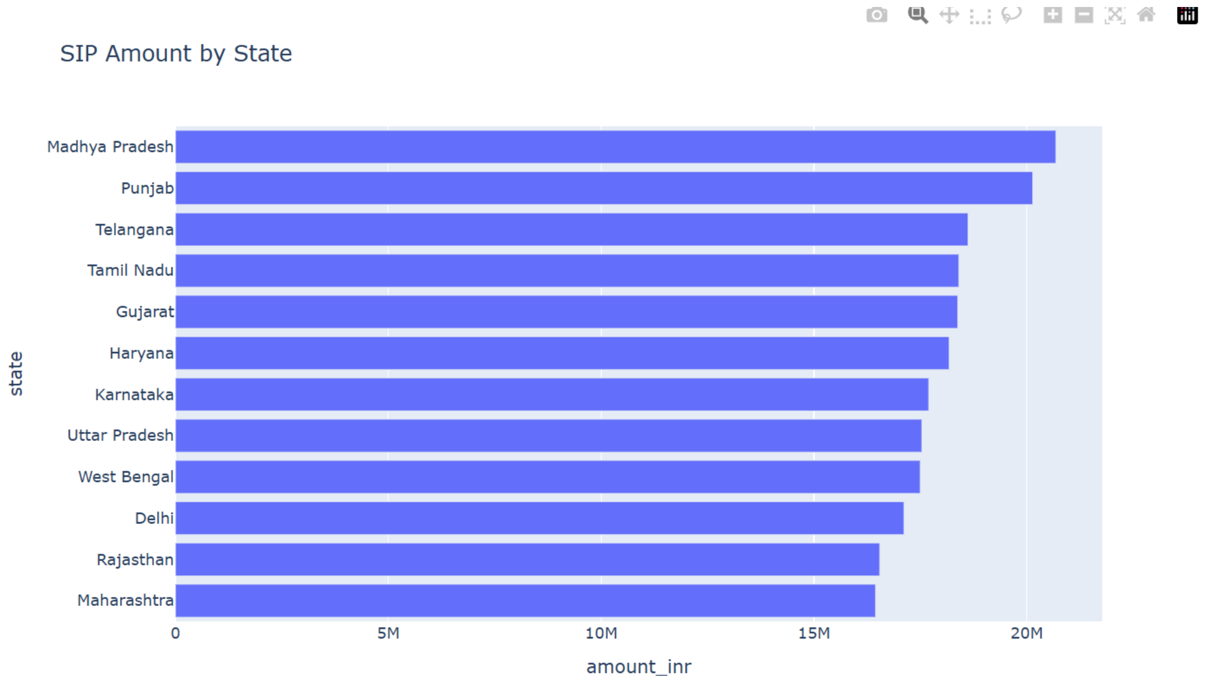
- Male investors represent the majority share.
- Female participation continues to grow.

Geographic Analysis

SIP Amount by State

Key findings:

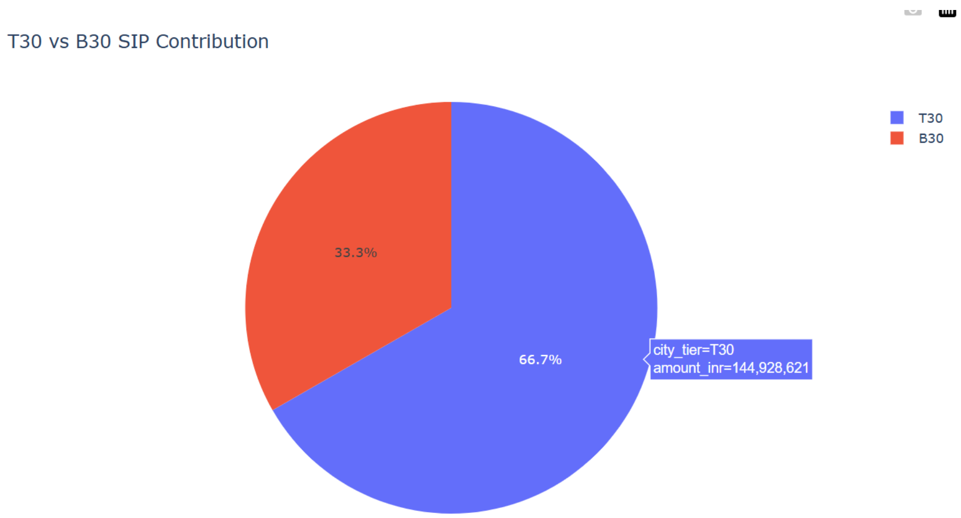
- Major metro states contribute most investments.
- Significant opportunity exists in underpenetrated regions.



T30 vs B30 Analysis

Key findings:

- T30 cities dominate investment activity.
- B30 growth indicates improving financial inclusion.



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6. Mutual Fund Performance Analysis

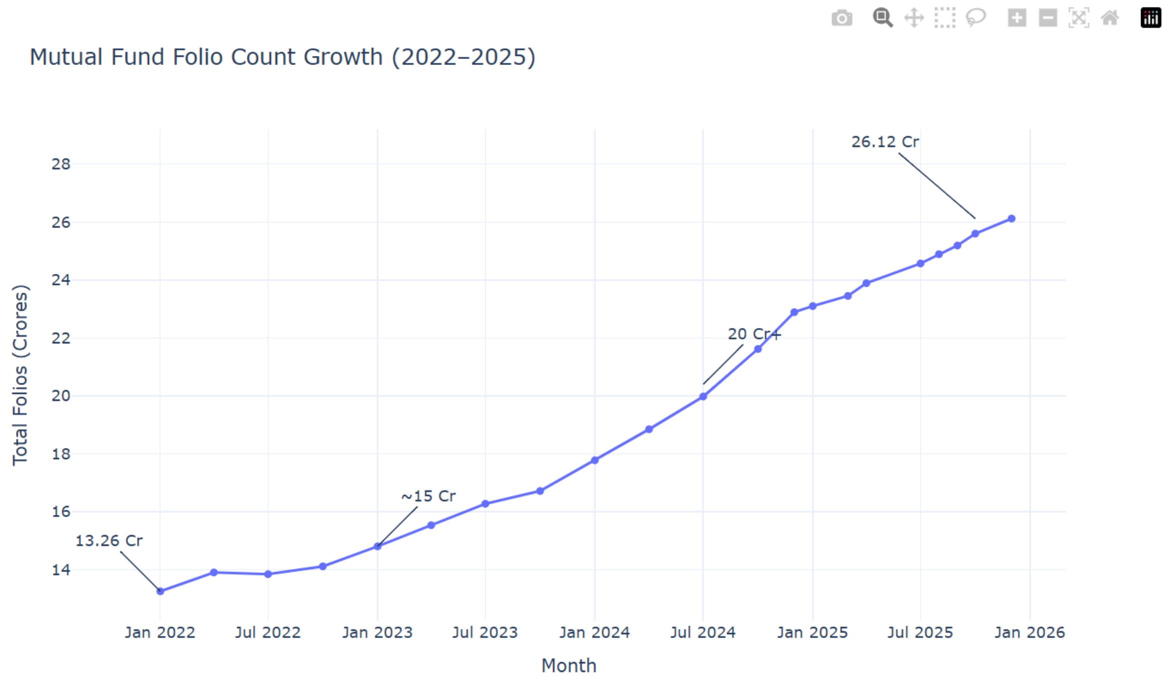
Return Analysis

Metrics evaluated:

- 1-Year Returns
- 3-Year Returns
- 5-Year Returns

Key findings:

- Large-cap funds demonstrated consistent performance.
- Some funds significantly outperformed benchmarks.



Risk-Adjusted Performance

Metrics:

- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta

Key findings:

- High Sharpe Ratio funds delivered superior risk-adjusted returns.
- Several funds generated positive alpha against benchmarks.

(Insert Risk Analysis Dashboard)

7. Risk Analytics

Historical Value-at-Risk (VaR)

Methodology:

- Daily return calculation
- 5th percentile estimation

Findings:

- Certain schemes exhibit significantly higher downside risk.
- Risk varies considerably across categories.

Conditional Value-at-Risk (CVaR)

Methodology:

- Average loss below VaR threshold

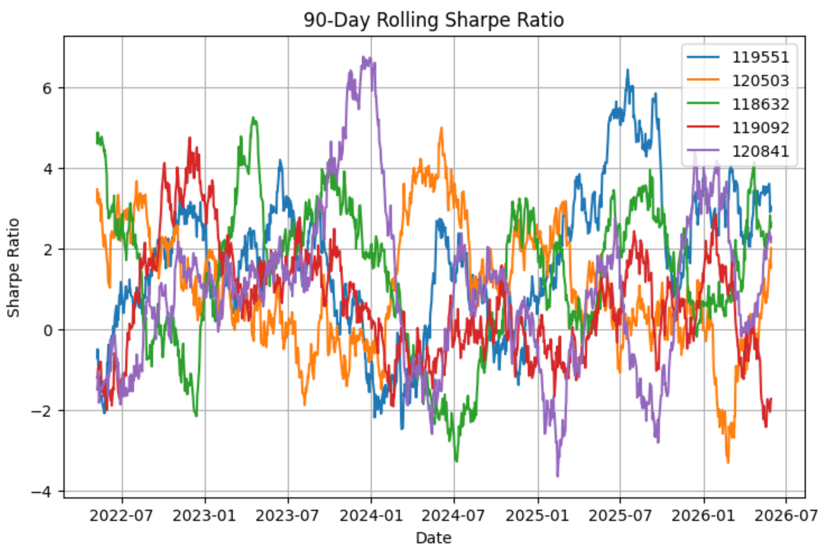
Findings:

- CVaR identified tail-risk exposure not visible through standard volatility measures.

Rolling 90-Day Sharpe Ratio

Findings:

- Performance stability differs across funds.
- Some funds maintain consistently superior risk-adjusted returns.



8. Investor Behaviour Analytics

Cohort Analysis

Findings:

- Newer investor cohorts invest larger amounts.
- Average SIP values have increased over time.

```
cohort_year,avg_sip_amount,total_invested,top_fund,transaction_count
2024,10996.885825361911,3491125187,148568,874
2025,13505.209580838324,30455243,119599,12
```

)

SIP Continuity Analysis

Findings:

- Most investors maintain regular SIP schedules.
- At-risk investors identified through transaction gap analysis.

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9. Portfolio Diversification Analysis

HHI Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to evaluate portfolio concentration.

Findings:

- Some equity funds exhibit concentrated sector exposure.
- Diversified funds generally display lower concentration risk.

```
ython scripts/sector_hhi_analysis.py

op 10 Most Concentrated Funds
    amfi_code       hhi       concentration_level
11     119092  0.296769       Highly Concentrated
30     148569  0.254992       Highly Concentrated
27     125498  0.253155       Highly Concentrated
5      102887  0.251383       Highly Concentrated
32     149323  0.241077   Moderately Concentrated
21     120505  0.238695   Moderately Concentrated
10     118635  0.237497   Moderately Concentrated
18     119599  0.232361   Moderately Concentrated
22     120506  0.231464   Moderately Concentrated
1      100033  0.227647   Moderately Concentrated

op 10 Most Diversified Funds
    amfi_code       hhi       concentration_level
0      100016  0.180588   Moderately Concentrated
2      101206  0.180042   Moderately Concentrated
13     119094  0.175100   Moderately Concentrated
26     125497  0.172411   Moderately Concentrated
24     120842  0.162062   Moderately Concentrated
9      118634  0.160299   Moderately Concentrated
14     119095  0.159582   Moderately Concentrated
15     119551  0.142491        Highly Diversified
25     120843  0.136206        Highly Diversified
6      102886  0.124020        Highly Diversified

aved to: data/processed/sector_hhi_analysis.csv
```

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10. Fund Recommendation Engine

A rule-based recommendation system was developed using:

- Risk Grade
- Sharpe Ratio
- Fund Performance Metrics

Risk Categories

- Low Risk
- Moderate Risk
- High Risk

The engine recommends the top-performing funds within each risk profile.

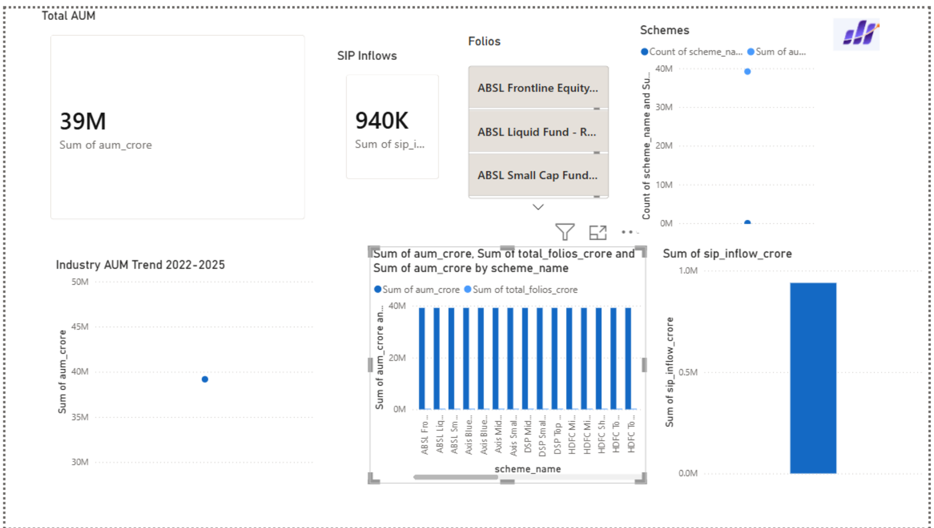
```
Enter Risk Appetite: high

Columns after merge:
['amfi_code', 'fund_house_x', 'scheme_name_x', 'category_x', 'sub_category', 'plan_
x', 'launch_date', 'benchmark', 'expense_ratio_pct_x', 'exit_load_pct', 'min_sip_am
ount', 'min_lumpsum_amount', 'fund_manager', 'risk_category', 'sebi_category_code',
 'scheme_name_y', 'fund_house_y', 'category_y', 'plan_y', 'return_1yr_pct', 'return
_3yr_pct', 'return_5yr_pct', 'benchmark_3yr_pct', 'alpha', 'beta', 'sharpe_ratio',
'sortino_ratio', 'std_dev_ann_pct', 'max_drawdown_pct', 'aum_crore', 'expense_ratio
_pct_y', 'morningstar_rating', 'risk_grade']

================================================================================
TOP 3 RECOMMENDED FUNDS (HIGH RISK)
================================================================================
 amfi_code                                    scheme_name          fund_house risk_c
ategory  sharpe_ratio  return_3yr_pct  expense_ratio_pct
    120842 Kotak Emerging Equity Fund - Regular - Growth   Kotak Mahindra MF
   High          0.96           18.23               1.56
    120505       ICICI Pru Midcap Fund - Regular - Growth ICICI Prudential MF
   High          0.95           18.08               1.36
    119598     SBI Small Cap Fund - Regular Plan - Growth     SBI Mutual Fund      Ve
ry High          0.94           23.39               1.43
```

11. Dashboard Overview

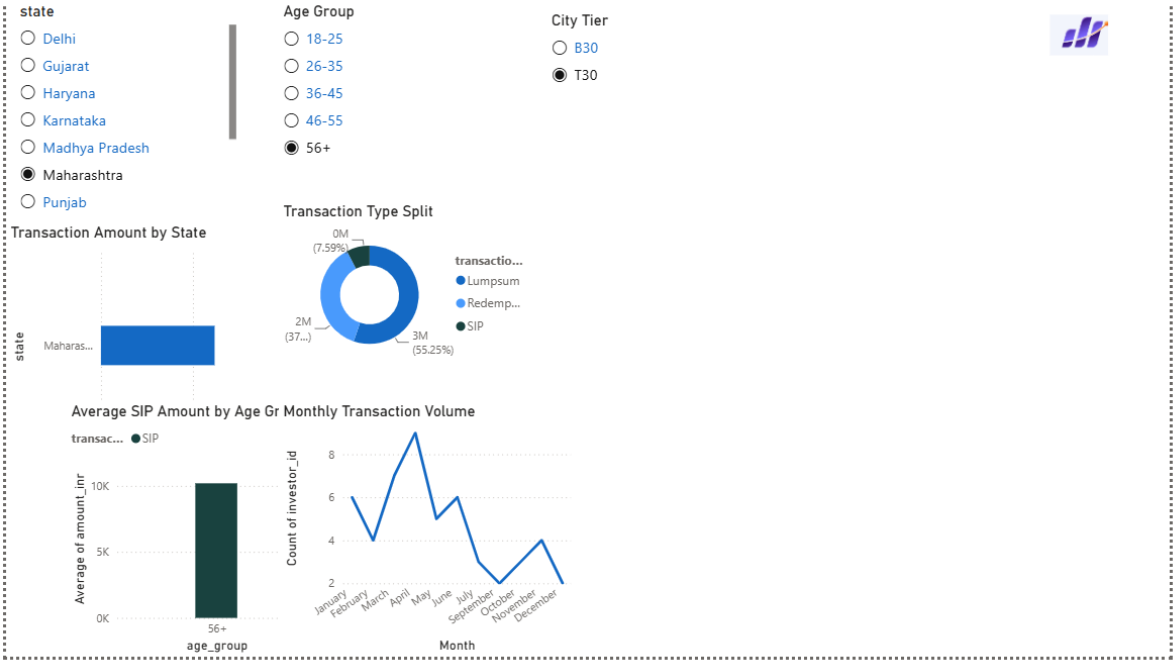
Dashboard Components

- Executive Summary
- Investor Analytics
- Fund Performance
- Risk Analytics
- Portfolio Concentration
- Geographic Analysis

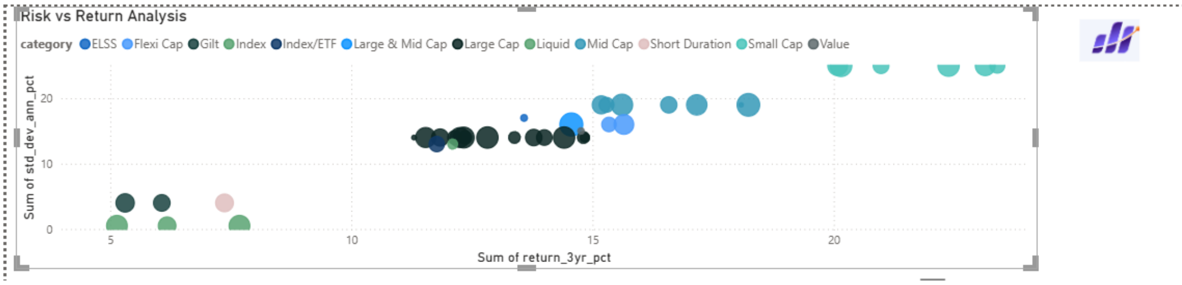
Screenshot 1: Overview Dashboard



Screenshot 2: Investor Analytics Dashboard



Screenshot 3: Risk Analytics Dashboard



Screenshot 4: Fund Performance Dashboard

scheme_name	fund_house	category	Sum of return_3yr_pct	Sum of sharpe_ratio	Sum of alpha	Sum of expense_ratio_pct	Sum of max_drawdov
UTI Nifty 50 Index Fund - Regular - Growth	UTI Mutual Fund	Index	12.10	0.93	0.93	1.57	
UTI Mid Cap Fund - Regular - Growth	UTI Mutual Fund	Mid Cap	15.61	0.82	1.12	1.51	
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	Flexi Cap	15.34	0.96	1.79	1.64	
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Small Cap	23.39	0.94	1.23	1.43	
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Small Cap	23.14	0.93	1.13	0.72	
SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	Gilt	6.07	1.52	1.60	0.77	
SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	Large Cap	12.36	0.88	0.87	1.54	
SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	Large Cap	11.30	0.81	1.78	0.66	
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	Small Cap	20.15	0.81	0.80	1.53	
Nippon India Large Cap Fund - Regular - Growth	Nippon India MF	Large Cap	14.00	1.00	0.86	1.51	
Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	Large Cap	12.33	0.88	1.70	0.72	
Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	Gilt	5.31	1.33	0.89	0.55	
Nippon India ETF Nifty 50 BeES	Nippon India MF	Index/ETF	11.77	0.91	1.80	0.89	
Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	ELSS	13.58	0.80	0.54	1.60	
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	14.81	1.06	1.62	1.46	
Mirae Asset Emerging Bluechip Fund - Regular - Growth	Mirae Asset MF	Large & Mid Cap	14.56	0.91	1.70	1.52	
Total			**563.56**	**54.47**	**50.14**	**49.48**	**-7**

2. Key Insights

Insight 1

Funds with the highest VaR values exhibited significantly larger downside exposure.

Insight 2

Recent investor cohorts contribute the largest investment volumes.

Insight 3

A small group of states account for the majority of SIP investments.

Insight 4

Higher Sharpe Ratio funds consistently outperform peers on a risk-adjusted basis.

Insight 5

Portfolio concentration varies significantly across equity schemes.

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13. Limitations

- Historical performance does not guarantee future returns.
- Limited availability of live market data.
- Simplified recommendation engine.
- Assumed transaction consistency for SIP analysis.
- Portfolio holdings snapshots may not represent real-time allocations.

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14. Recommendations

For Investors

- Focus on risk-adjusted returns rather than absolute returns.
- Maintain SIP discipline for long-term wealth creation.
- Diversify across categories and sectors.

For Fund Houses

- Improve investor engagement in B30 regions.
- Monitor SIP continuity to reduce churn.
- Promote diversified investment strategies.

For Future Enhancements

- Machine Learning-based recommendation engine.

- Real-time NAV integration.
- Portfolio optimization models.
- Predictive investor churn analytics.
- Automated dashboard refresh pipelines.

Conclusion

The Mutual Fund Analytics Platform successfully combines data engineering, risk analytics, business intelligence, and investment analysis into a unified solution. The project demonstrates practical application of Python, SQL, Power BI, and financial analytics concepts while delivering meaningful insights into mutual fund performance and investor behaviour.